

LakeheadLMS

CS5450 Mobile Programming
Challenge 2: React Native Mobile Mini Portal
Group #1: Education / LMS Portal

App Name	LakeheadLMS
Tagline	Your Learning. Your Campus. Your Portal.
GitHub	https://github.com/omavaiya4/LakeheadLMS
Firebase	lakehead-lms (Firestore + Email/Password Auth)
Android Pkg	ca.lakeheadu.lms

Group Members

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Project Overview

LakeheadLMS is a full-featured Education/LMS mobile portal built with React Native (Expo) and Firebase. The app supports two user roles — **Student** and **Instructor** — each with a personalized dashboard, course management, grade tracking, announcement posting, powerful search, and secure Firebase authentication. All data is persisted in Cloud Firestore.

Feature Summary

- Firebase-initialized React Native (Expo) app with a guarded navigation path.
- Professional Lakehead-branded visual design with deep navy, white, crimson red, and gold accent colors.
- Responsive mobile layout with a persistent bottom navigation bar: Home, Courses, Grades, Search, Profile.
- Firestore-backed courses, grades, announcements, and user profiles.
- Role-based personalized dashboard — students see enrolled courses and GPA; instructors see teaching courses.

- Course management — instructors create courses; students browse and enroll from the available list.
- Grade tracker — instructors enter grades (assignment / quiz / midterm / final); students view letter grades and GPA.
- Announcements — instructors post per-course announcements; students view them in the course detail screen.
- Global search across courses, students, and announcements with filter chip selector.
- Profile screen with avatar, role badge, and inline editing for name and department.
- Consultation form placeholder — Contact tab ready for extension.
- Valid user registrations and grade entries are written to Firestore with server timestamps.

Firestore Configuration

This project uses:

- firebase/auth — for Email/Password user authentication and session management.
- firebase/firestore — for all portal data reads and writes.

Configured Firebase project details:

Firebase Project ID	lakehead-lms (replace with your own)
Authentication	Email / Password sign-in provider
Firestore Database	(default)
Firestore Region	northamerica-northeast1 (or nearest)
Android Package	ca.lakeheadu.lms

The app initializes Firebase in **constants/firebase.ts**:

```
import { initializeApp } from 'firebase/app';
import { getAuth } from 'firebase/auth';
import { getFirestore } from 'firebase/firestore';
const app = initializeApp(firebaseConfig);
export const auth = getAuth(app);
export const db = getFirestore(app);
```

The real Firebase config values are placed directly in **constants/firebase.ts**. This file should be added to **.gitignore** to avoid exposing API keys in public repositories. A **firebase.ts.example** file with placeholder values is committed for documentation.

Firestore Data Model

Collection	Purpose
users	uid, email, name, role (student instructor), studentId, department, enrolledCourses[]
courses	title, code, credits, description, schedule, room, instructor, instructorId, enrolledStudents[]
grades	courseCode, courseTitle, assignmentName, score, maxScore, type, studentId, studentName, instructorId, createdAt
announcements	title, body, courseId, courseCode, instructorId, createdAt

The **users**, **courses**, and **announcements** collections are readable by authenticated users. The **grades** collection is restricted so students see only their own records and instructors see only grades they have entered. Firestore security rules should be deployed before going to production.

Setup Instructions

Install prerequisites:

- Node.js 18+ — <https://nodejs.org>
- Android Studio with Android SDK and at least one Android Virtual Device (AVD) configured
- Expo CLI — `npm install -g expo-cli`
- A Firebase account and project at <https://console.firebase.google.com>

Verify Node and Expo:

```
node --version
npx expo --version
```

Enable Firebase services:

- Go to Firebase Console → Authentication → Get started → Enable Email/Password.
- Go to Firebase Console → Firestore Database → Create database (start in test mode).
- Click the Web icon → Register app → copy the firebaseConfig object.
- Paste the config into constants/firebase.ts replacing the placeholder values.

Run Instructions

Install dependencies:

```
npm install
```

Start the Expo development server:

```
npx expo start
```

Run on Android emulator (press **a** in the terminal after the server starts):

```
# In the Expo terminal, press: a
# OR run directly:
npx expo start --android
```

Optional — run on a physical Android device with Expo Go (+2 bonus marks):

- Install the Expo Go app from the Google Play Store.
- Connect your phone to the same Wi-Fi network as your laptop.
- Scan the QR code shown in the terminal with the Expo Go app.

Optional Chrome/web responsive check:

```
npx expo start --web
```

Android emulator execution is the primary demonstration path for this submission. Chrome/web is included only as an additional platform check.

Project Structure

```
LakeheadLMS/
|-- app.json # Expo app configuration
```

```

|-- package.json # npm dependencies
|-- tsconfig.json # TypeScript configuration
|-- babel.config.js # Babel + Reanimated plugin
|-- README.pdf # This file
|-- constants/
| |-- Colors.ts # App-wide color palette (Lakehead theme)
| `-- firebase.ts # Firebase app initialization + exports
|-- hooks/
| `-- useAuth.tsx # Auth context: login, register, logout, profile state
|-- components/
| `-- BottomNav.tsx # Persistent bottom navigation bar (5 tabs)
|-- app/
| |-- _layout.tsx # Root layout, AuthProvider, navigation guard
| |-- auth/
| | |-- login.tsx # Email + password login screen
| | `-- register.tsx # Registration with role selector (Student / Instructor)
| |-- tabs/
| | |-- dashboard.tsx # Personalized home dashboard
| | |-- courses.tsx # Course list, create (instructor), enroll (student)
| | |-- grades.tsx # Grade entry (instructor) + GPA view (student)
| | |-- search.tsx # Global search with filter chips
| | `-- profile.tsx # User profile view and inline editing
| `-- course/
| `-- [id].tsx # Course detail: overview / announcements / students
`-- assets/ # App icon, splash screen images

```

Validation Results

Final validation completed successfully:

- npm install passed — all dependencies resolved without errors.
- npx expo start launched successfully and served the development bundle.
- Android emulator run was tested successfully after Firebase configuration.
- User registration (Student and Instructor roles) verified against Firebase Auth and Firestore users collection.
- Course creation by instructor and student enrollment verified against Firestore courses collection.
- Grade entry by instructor verified against Firestore grades collection; student GPA view confirmed.
- Announcement posting by instructor verified; announcements appear in course detail screen for students.
- Global search returns results across courses, students, and announcements collections.
- Profile editing updates Firestore users document correctly.
- Navigation guard redirects unauthenticated users to the login screen.

Demo Flow

Recommended Zoom demonstration:

- (1) Open the app on an Android emulator and confirm the Login screen loads.
- (2) Register a new Instructor account — select the Instructor role, fill in name and email.
- (3) On the Dashboard, confirm the instructor sees 'My Teaching Courses' and quick action cards.
- (4) Navigate to Courses and create a new course (e.g. COMP5413 Advanced Data Structures).

- (5) Log out and register a new Student account — select the Student role with a student ID.
- (6) On the Student Dashboard, navigate to Courses → Enroll in the course just created.
- (7) Navigate to Grades — confirm the student sees an empty grade tracker.
- (8) Log back in as the Instructor, go to Grades → Add Grade, enter a score for the student.
- (9) Log back in as the Student, confirm the grade now appears with the correct letter grade and average.
- (10) Navigate to the course detail screen → Announcements tab → confirm instructor-posted announcements appear.
- (11) Navigate to Search, type a course code, confirm search results appear with correct filter chips.
- (12) Open Profile, tap the pencil icon, update the name, and confirm the change saves to Firestore.

ZIP Packaging Instructions

Include in the submission ZIP:

- app/
- components/
- constants/
- hooks/
- assets/
- app.json
- package.json
- tsconfig.json
- babel.config.js
- README.pdf
- README.md (optional)

Do **not** include:

- node_modules/
- .expo/
- android/ or ios/ build artifacts
- constants/firebase.ts (if it contains real API keys — add to .gitignore)
- Any private Firebase service account JSON files

Known Limitations

No build-blocking limitations are known for the Android emulator demonstration. The app is configured for Android through Expo's managed workflow. iOS has not been tested but may work with minor adjustments. Firestore security rules are currently in test mode — production deployments should restrict read/write access per user role before public release.